ANKIT KUMAR | 25AG62R05

M.Tech in LAND AND WATER RESOURCES ENGINEERING, IIT Kharagpur

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.Tech, Land & Water Resources Engineering	IIT Kharagpur	7.15 / 10
2025	B.Tech, Agricultural Engineering	Anand Agricultural University	8.049 / 10
2021	Intermediate Annual Exam	BSEB	83.8%
2019	Secondary School Examination	CBSE	89.4%

INTERNSHIPS

Research Intern (Deep Learning & Generative AI) | IITM, Ministry of Earth Sciences, Pune *[May'26-Jul'26]*

- Worked on 3 studies on Arka HPC: INSAT-3DR CREF retrieval (U-Net, GAN, Transformer), physics-informed retrieval with diffusion downscaling, and **NIRA**, a near real-time INSAT-radar dashboard.
- Built a pipeline with uncertainty quantification; **deployed a FastAPI + React dashboard** for Northern and Central India; ran explainability analyses and a pre-registered experiment.
- Showed the soft CSI used for CREF retrieval is the α=β=1 case of the **Tversky family**; mapped (α, β) with CIs (doubling the FP penalty: CSI 0.217).
- Tested diffusion downscaling vs bicubic (FSS@35 0.862) and spectral baselines: no accuracy claim, but **25.7 cells/scene vs 25.0 true**.

Research Intern, ML & Applied AI | EXPERIQS Pvt. Ltd. (IIT Bombay-based startup), Remote *[Jan'26-Jun'26]*

- Built **GeoCopilot**, a guardrailed LLM tool-calling agent orchestrating a 9-module climate-risk platform (multi-agent orchestration).
- Deployed a conversational agent over live CMIP6 / ERA5 / NASA POWER feeds; integrated an **LLM evaluation framework** for factual grounding.
- Supported flood-risk and soil-property platforms (ML, Sentinel-1/2, Google Earth Engine) and physical climate risk assessments.
- Upper Beas flood study (CDRI Fellowship 2025-26): supported **IMERG-forced HEC-HMS + HEC-RAS 2D** modelling of the Jul-2023 flood: 23.07 km² inundated, 77.6% Extreme Hazard.
- Contributed an 11-factor **AHP susceptibility screen** and road-passability analysis: **322 of 1,534 road segments** impassable to cars, 318 to ambulances; manuscript under revision.
- Ran climate tests on HEC-HMS (75 sub-basins, 8,847 m³/s peak): IMERG-anchored delta change, 4 CMIP6 models; SSP5-8.5 far-future peak **+22.3%**.

PROJECTS

Parishkar | Autonomous Software Engineering Agent for SWE-Bench with LangGraph and MCP | Self Project | GitHub *[Jul'26-Aug'26]*

Python | LangGraph | Model Context Protocol (MCP) | Docker | Multi-Agent Orchestration | LLM Evaluation

- Built an autonomous coding agent (**LangGraph, 11 lifecycle hooks**) with an **MCP tool server** exposing 5 sandboxed, path-contained, allowlisted tools.
- Designed and evaluated the agent across **40 SWE-bench Lite tasks**, comparing single-shot, retry-loop and test-driven retry strategies.

HydroSent | Climate Risk & Resilience Platform | Prof. Amey Pathak | GitHub *[Mar'26-May'26]*

FastAPI | Next.js 14 | Google Earth Engine | Leaflet | Docker | Llama LLM Copilot

- Built a climate-risk dashboard for flood, drought, groundwater, weather and infrastructure risk via **GEE-powered ML pipelines**.
- Architected a **FastAPI backend**, Next.js 14 frontend and Leaflet maps (Dockerised); integrated a **Llama LLM copilot** and an operations-research module.

RankLab | E-commerce Search Ranking with LLM Query Rewriting and Relevance Judging | Self Project | GitHub *[May'26-Jun'26]*

Python | LightGBM | LambdaMART | Dense Embeddings (FAISS) | Groq LLM | Streamlit

- Built an end-to-end search pipeline: BM25, dense retrieval (HNSW, **4.5 ms latency**, bge-base-en-v1.5), LambdaMART; **NDCG@10 0.365**.
- LLM query rewriting (Groq gpt-oss-120b) gave a **+1.3% BM25 recall lift** (95% CI excluding zero); LLM relevance judging reached Cohen's κ = 0.478.
- Built a synthetic A/B/n experiment simulator with **CUPED variance reduction** (~15-16% cut) and O'Brien-Fleming sequential testing.

TerrAI | Soil Intelligence Platform (GeoAI) | Self Project | GitHub *[Apr'26-Jun'26]*

Python | PyTorch | FastAPI | Next.js | Docker | Google Cloud Run

- Built a platform predicting soil properties from satellite, weather and terrain data; benchmarked **13+ models** (ML, MLP, deep kriging, spatial CNNs).
- Implemented **spatial block CV** and Area-of-Applicability; integrated frozen Prithvi-EO ViT with SHAP and calibrated uncertainty.

NIRMEGH | Cloud Removal for LISS-IV Imagery over North-East India | BAH 2026 Finalist | GitHub *[Jul'26-Aug'26]*

Python | PyTorch | NAFNet | Diffusion Sampler | LISS-IV MX | Residual Learning

- Built a **NAFNet model** (29.2M params): **+3.01 dB** on cloud pixels (34.13 vs 31.12 dB); clear-ground 33.63 → 34.55 dB, 332 held-out patches.
- Built a perceptual ensemble with a mean-reverting diffusion sampler (FiLM, EMA), trading 0.74 dB for a 13.7× closer output distribution.

CREF Retrieval from INSAT-3DR Imagery | Physics-Informed Deep Learning with Diffusion Downscaling to 1 km | IITM, Pune *[May'26-Jul'26]*

Python | PyTorch | U-Net | GAN | Transformer | Diffusion Models | Conformal Risk Control | First-author manuscript

- Benchmarked U-Net, GAN-refined, Transformer on 9-channel INSAT-3DR: R 0.766-0.783 vs pixelwise 0.242; GAN: CSI +8.6%, false alarms -13.7%.
- Built a physics-informed U-Net with a differentiable Tversky-generalised objective over a 440 km domain (4 monsoons): CSI@35 = 0.46 at POD 0.65; objective +0.0659 CSI vs +0.0017 for 12 physics channels (~39×).
- Applied **conformal risk control** (miss-rate bound held in 12/12 cells).

AWARDS AND ACHIEVEMENTS

- **Grand Finalist (Top 34 of 15,104 teams)** | Bharatiya Antariksh Hackathon 2026, ISRO; 2 teams shortlisted (cloud removal and urban heat mitigation)
- **National Finalist (Top 38)** | MoPR AI/ML Hackathon, IIT Tirupati Navavishkar i-Hub Foundation

SKILLS AND EXPERTISE

Programming & MLOps: Python, SQL, TypeScript (Next.js), Git, FastAPI, Asyncio, Docker, Kubernetes, MLflow, Prometheus, Google Cloud Run
GenAI: LLMs, RAG, tool calling, LangGraph, MCP, multi-agent orchestration, LLM evaluation, LoRA/QLoRA, Groq / Llama APIs
Retrieval: BM25, FAISS/HNSW, embeddings, reranking, LambdaMART, pgvector, NDCG evaluation
ML/DL: PyTorch, scikit-learn, XGBoost, LightGBM, Transformers, CNNs, U-Net, GAN, diffusion models, SHAP, spatial cross-validation

CERTIFICATIONS

- **IIRS-ISRO:** AI/ML for Geodata Analysis | **Educative:** ML with NumPy, pandas, scikit-learn | **Univ. of Alaska Fairbanks:** GIS Essentials

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant**, Lab Geoinformatics (AG69039, 2026) | **Teaching Assistant**, Advanced Remote Sensing & Geospatial Intelligence for Defence (Geospatial Academy, IIT Kharagpur, Aug-Sep'26) | **CDC Head**, AFES Society | **PG Representative**, E-Cell, IIT Kharagpur